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Measuring Security in Real Time: The Haiti Security Index

A Text-Based Monthly Measure of Violence and Threat Perceptions

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Abstract

This paper introduces the **Haiti Security Index (HSI)**, a monthly text-based measure of domestic security conditions spanning January 1997 to March 2026 (351 months). The HSI is built from systematic keyword searches of English- and French-language news articles from nine outlets—four Haitian and five international—aggregated and normalized by total media volume. Following the methodology of Caldara and Iacoviello (2022) Geopolitical Risk Index, the HSI separately tracks *realized violent acts* (HSI-Acts) and *forward-looking threat perceptions* (HSI-Threats), as well as a composite measure (HSI-All), each normalized to a sample mean of 100. The index successfully captures all major security episodes in Haiti's modern history: the 2004 political crisis (annual average 219), the 2010 earthquake (112), the 2021 presidential assassination and second earthquake (231), and the 2022–2025 gang consolidation period (143–215). A structural break emerges after 2017, when threat perceptions systematically exceed realized violence—a regime shift consistent with intimidation-led insecurity. The HSI fills a critical data gap in IMF surveillance by providing a high-frequency, replicable, and forward-looking security indicator suitable for nowcasting GDP, inflation, exchange rate pressures, and other macroeconomic outcomes.

JEL Classification: D74, E01, O17, C82

Keywords: Haiti, security index, text analysis, geopolitical risk, gang violence, nowcasting, conflict measurement

Contents

Introduction

Haiti occupies a singular position in the Western Hemisphere: it is the poorest country in the region and among the most affected by chronic insecurity. Gang violence, kidnappings, territorial disputes, political assassinations, and periodic humanitarian catastrophes have punctuated its modern history, directly shaping macroeconomic outcomes. Port blockades disrupt fuel and food imports. Armed checkpoints reduce labour mobility and firm activity. Assassination risk reduces investment and deepens institutional fragility. Fear of violence alters household consumption and savings behaviour even when direct confrontation is absent. Despite the centrality of insecurity to Haiti's economic trajectory, no systematic high-frequency measure of its security environment existed before this work.

This paper fills that gap. The **Haiti Security Index (HSI)** applies a transparent text-analysis methodology—adapted from the Geopolitical Risk (GPR) index of Caldara and Iacoviello (2022)—to construct a monthly measure of domestic security conditions from January 1997 through March 2026. The index draws on over 110,000 security-related news articles drawn from a corpus of up to 463,000 articles per month, sourced from nine outlets in English and French. It separately quantifies *realized violence* (HSI-Acts) and *forward-looking threat perceptions* (HSI-Threats), enabling analysis of both acute shocks and the expectational channels through which anticipated insecurity affects economic behaviour.

The HSI makes three substantive contributions. First, it provides a replicable, high-frequency indicator for a country where standard administrative data—national accounts, trade statistics, household surveys—are often delayed, incomplete, or unavailable. The monthly cadence and near-real-time availability make it directly useful for IMF Article IV consultations, Staff-Monitored Program (SMP) reviews, and nowcasting exercises. Second, the decomposition into acts and threats enables analysis of distinct macroeconomic channels. HSI-Threats captures the fear-driven dimension of insecurity—precautionary FX demand, preemptive price-setting, investment withdrawal—that operates through expectations before violence materialises. Third, the historical span of nearly three decades allows systematic identification of structural changes in Haiti's security regime, most notably a post-2017 shift toward a threat-dominated environment in which anticipated insecurity persistently exceeds realized violence.

The HSI offers several concrete advantages over existing security data for Haiti. The most systematic alternative is the BINUH Security and Crime Statistics dataset published by the UN Peace and Security Data Hub, which covers monthly homicide and kidnapping counts from July 2018 onward. While valuable, the BINUH data have three important limitations relative to the HSI. First, the *time horizon* of BINUH is short—roughly seven years—whereas the HSI spans nearly three decades (1997–2025), enabling analysis of long-run structural changes, regime shifts, and historically unprecedented episodes such as the 2010 earthquake and the 2004 political crisis. Second, the HSI captures a broader *coverage* of security conditions: it encompasses not only realized violence (homicides, kidnappings) but also threat perceptions, political insecurity, and institutional fragility, which are absent from administrative crime statistics. Third, the *near-real-time* availability of the HSI contrasts with the reporting lags that often affect field-collected administrative data in fragile states. These advantages are substantiated empirically in Section 7, where the HSI-All index is shown to correlate strongly with BINUH crime indicators over the 2018–2025 period of overlap, validating the text-based methodology against independent ground-truth data.

The index is constructed using the Factiva platform, which provides systematic access to print and digital archives from Haitian, regional, and international news outlets. Security-related articles are identified using a bilingual keyword dictionary grouped into "acts" terms (assault, killing, kidnapping, blockade), "threats" terms (warning, risk, curfew, state of emergency), and institutional vocabulary (gang, police, UN mission). Each article count is divided by the total number of articles from the same outlets in the same month, scaling by media volume to ensure that index movements reflect the prominence of security concerns rather than changes in overall publication rates. The resulting shares are normalized to 100 over the full sample.

Figure 1 presents the full HSI-All series alongside its 12-month moving average from 1997 to 2025. Figure 2 overlays HSI-Acts and HSI-Threats with event-category shading, revealing episodes where threat perceptions diverge systematically from realised violence.

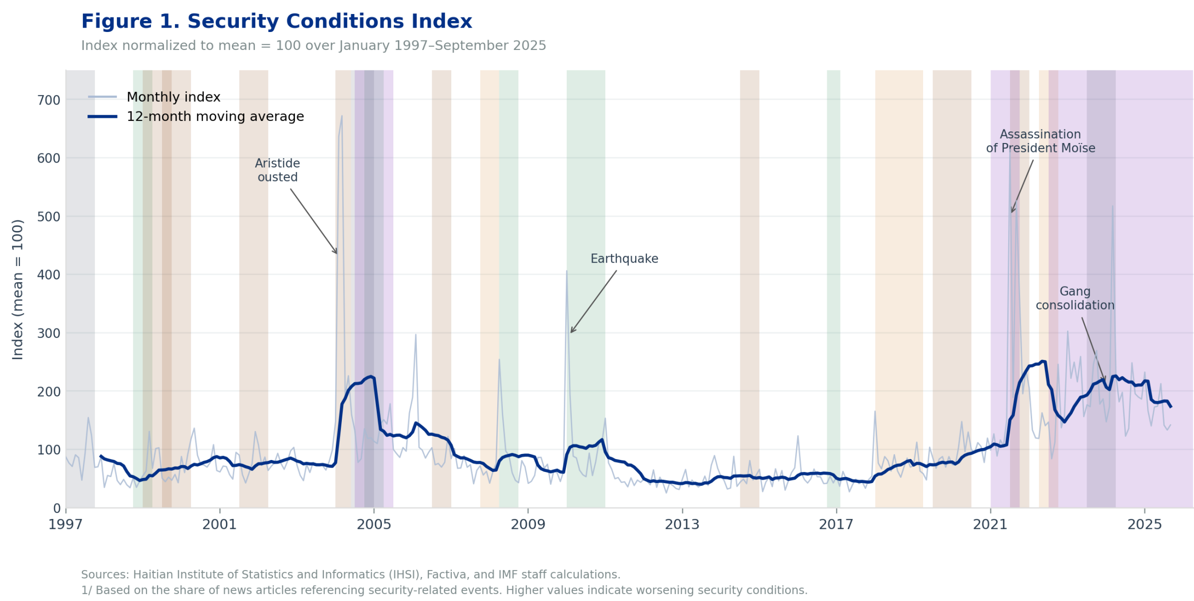

Figure 1: Security Conditions Index, January 1997–November 2025
Sources: Haitian Institute of Statistics and Informatics (IHSI), Factiva, and IMF staff calculations. 1/ Based on the share of news articles referencing security-related events. Higher values indicate worsening security conditions.

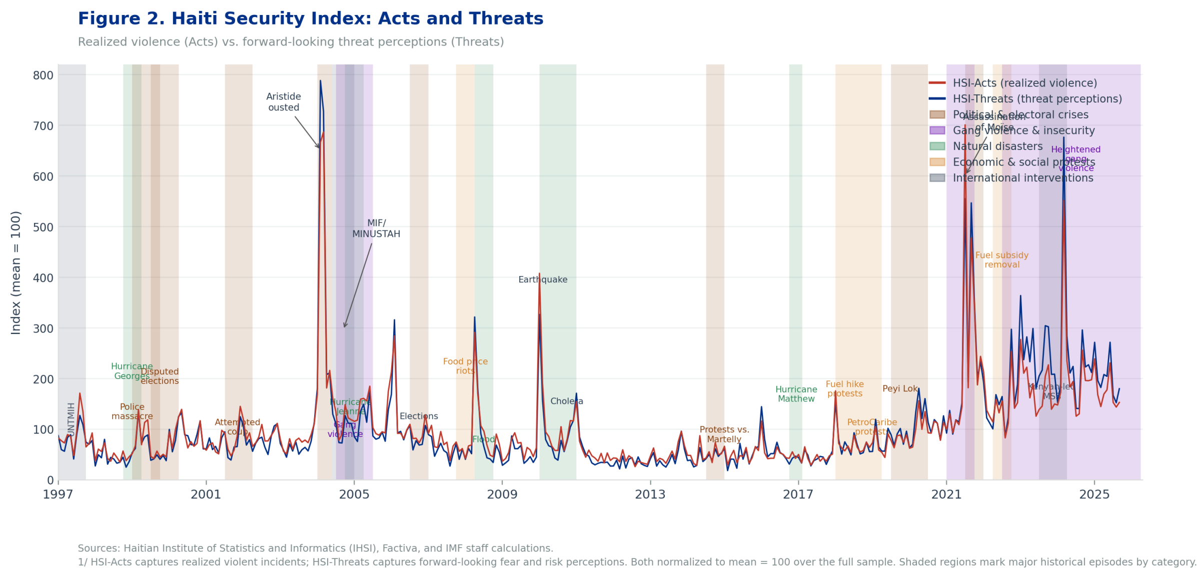

Figure 2: Haiti Security Index: Threats and Acts, January 1997–November 2025
Sources: IHSI, Factiva, and IMF staff calculations. Shaded regions denote major event categories (see legend). Higher values indicate greater security pressure.

The remainder of the paper is organized as follows. Section 2 reviews related literature. Section 3 describes the data and sources. Section 4 presents the methodology. Section 5 documents the statistical properties and major episodes. Section 6 discusses macroeconomic applications. Section 7 validates the HSI against independent crime statistics from BINUH. Section 8 concludes.

Related Literature

This paper sits at the intersection of three strands of research: text-based macroeconomic indicators, the economic consequences of conflict and insecurity, and real-time measurement for surveillance purposes.

Text-Based Macroeconomic Indicators

The use of news text to construct economic indicators has grown rapidly since Baker et al. (2016) introduced the Economic Policy Uncertainty (EPU) index. The EPU counts newspaper references to policy uncertainty, normalized by total articles, and has been constructed for over 40 countries. Caldara and Iacoviello (2022) extend this approach to geopolitical risk, building a monthly index from 1900 for the United States and showing that geopolitical risk shocks reduce investment, employment, and trade. The HSI follows their methodology closely but adapts it for a small, fragile economy where domestic security—rather than global geopolitics—is the binding constraint on macroeconomic activity.

Text-based methods have been applied to measure financial sentiment, firm-level fundamentals, and regulatory uncertainty (Tetlock et al., 2008; Loughran and McDonald, 2011). A common concern is that news coverage reflects editorial choices and media incentives rather than underlying conditions. The HSI addresses this by (i) normalizing by total media volume to control for aggregate reporting cycles, (ii) drawing on multiple outlets with distinct editorial orientations, and (iii) restricting coverage to a tightly defined keyword dictionary with proximity filters that reduce false positives.

Conflict, Insecurity, and Economic Outcomes

A large empirical literature documents the negative economic effects of conflict and insecurity. Acemoglu et al. (2008) and Besley and Persson (2011) show that state fragility and violence reduce investment and long-run growth. Amodio and Di Maio (2018) document how conflict disrupts input markets and depresses firm productivity. For Haiti specifically, existing work documents the effects of individual episodes—the 2010 earthquake, political crises—but lacks a continuous high-frequency indicator that would allow systematic estimation of the security–economy relationship.

Existing conflict databases such as ACLED and UCDP offer event-level data but are designed for cross-country research rather than macroeconomic surveillance: they record discrete events rather than providing a continuous intensity measure, and their release schedules do not align with monthly monitoring needs. The HSI complements these datasets by providing a continuous, normalized monthly series that captures both the incidence and the perceived intensity of insecurity.

Real-Time Indicators for Fragile States

Standard macroeconomic surveillance relies on administrative data that are often unavailable or severely delayed for fragile and conflict-affected states. Haiti's national accounts are published with lags of 12–18 months; consumer price data are irregular; and external sector statistics are frequently revised. This creates a case for "real-time" proxies drawn from alternative data sources—satellite imagery, mobile data, nighttime lights, and media text—to bridge informational gaps. The HSI contributes to this toolkit by providing a replicable, automatable monthly indicator that can be updated as new articles enter the Factiva archive.

Data

News Sources

The HSI draws on nine news outlets selected to balance depth of local coverage, language diversity, and continuity of digital archives. Four Haitian outlets anchor the index with granular event-level reporting:

- **Le Nouvelliste** (est. 1898): Haiti's oldest continuously published newspaper. French-language. Provides the most comprehensive domestic archive.

- **Gazette Haiti**: French-language daily with strong coverage of security events in the capital and provinces.
- **Haiti Libre**: Bilingual French–English outlet with accessible digital archives.
- **Haitian Times**: English-language, diaspora-oriented, with consistent Haiti-focused coverage.

Five international and regional outlets provide broader context, ensure resilience when domestic media face operational disruptions, and capture international perception of Haiti's security environment:

- Agence France-Presse (AFP); Reuters; Associated Press (AP)
- CE NoticiasFinancieras (regional newswire)
- Miami Herald; BBC; *The New York Times*; *The Washington Post*; The Canadian Press

All articles are sourced through Factiva, which provides structured metadata (publication date, outlet, word count) and full-text search with proximity operators. The total corpus spans 1977 through April 2026 and ranges from approximately 2,000 articles per month in the late 1970s to over 463,000 articles per month by 2024. For the HSI proper, the sample is restricted to January 1997 onward to ensure consistent digital coverage across all nine sources.

Sample Coverage

The estimation sample covers 351 months from January 1997 to March 2026. Over this period, the corpus averages 216,893 articles per month, with a minimum of 91,636 and a maximum of 463,141. Security-related articles average 189 per month for English-language sources and 30 per month for French-language sources, with combined totals averaging 219 per month. English-language outlets account for approximately 97 percent of matched security articles, reflecting the larger digital archives of international wire services.

Methodology

Conceptual Framework

The HSI conceptualises security risk as the prominence of security-related reporting in the news, conditional on total media volume. This approach rests on two assumptions: (i) journalists and editors systematically increase coverage of security topics when conditions deteriorate, and (ii) the ratio of security coverage to total coverage is a better proxy for the prevalence of insecurity than raw article counts, since raw counts are contaminated by changes in total publication rates.

The index distinguishes two conceptually distinct dimensions of insecurity. *Realized violent acts* (HSI-Acts) are reports of events that have already occurred—killings, kidnappings, clashes, blockades. They capture current conditions and generate immediate economic disruptions. *Threat perceptions* (HSI-Threats) capture forward-looking signals: warnings, risk assessments, declarations of emergency, and anticipatory language. They are particularly relevant for macroeconomic analysis because expectations of insecurity shape behaviour in advance of realized violence: firms preemptively raise prices, households seek FX, investors postpone commitments.

Keyword Dictionary

Articles are identified using a bilingual keyword dictionary applied through the Factiva search interface. The base filter requires any article to reference Haiti or major Haitian locations within 50 words of a security-related keyword (the `NEAR/50` operator):

```
(Haiti OR Haitian OR "Port-au-Prince" OR "Cap-Haitien" OR
Gonaives OR Artibonite) NEAR/50 (<security keyword>)
```

Exclusion filters remove sports, entertainment, metaphorical language, and natural disaster coverage. The dictionary is grouped into Acts terms, Threats terms, and Institutional vocabulary. See Appendix Table 5 for the complete list.

Index Construction

The index is constructed in four steps.

Step 1: Monthly article counts. For each month t and sub-index $k \in \{\text{All, Acts, Threats}\}$, count the number of articles matching the relevant keyword group:

$$\text{Count}_t^{(k)} = \text{number of articles in month } t \text{ matching keyword group } k.$$

Step 2: Normalize by total media volume. Divide each count by the total number of articles from all outlets in month t and multiply by 100:

$$\text{Share}_t^{(k)} = \frac{\text{Count}_t^{(k)}}{\text{TotalArticles}_t} \times 100.$$

Step 3: Sample restriction. Retain observations from January 1997 onward to ensure consistent digital coverage across all nine source outlets.

Step 4: Mean normalization. Divide each monthly share by the full-sample mean share and multiply by 100, so the index has a mean of 100 over the estimation sample:

$$\text{HSI}_t^{(k)} = \frac{\text{Share}_t^{(k)}}{\overline{\text{Share}}^{(k)}} \times 100,$$

where $\overline{\text{Share}}^{(k)}$ is the simple mean of $\text{Share}_t^{(k)}$ over all $T = 351$ months.

Interpretation

The HSI is an intensity measure, not a count of events. A value of 100 corresponds to average security conditions over the full 1997–2026 sample. Values above 100 indicate above-average security pressure; values below 100 indicate below-average pressure. For medium-term trend analysis, 12-month moving averages (HSI_MA12) are also computed to smooth high-frequency noise.

Properties of the Haiti Security Index

Descriptive Statistics

Table 1 reports summary statistics for the three HSI series and the total article corpus over the full 1997–2026 sample. All three sub-indices display substantial right skew, with maximum values between five and seven times the sample mean, reflecting the episodic nature of security crises.

Table 1: Summary Statistics, January 1997–November 2025 ($N = 347$ months)

Series	Mean	Std. Dev.	Min	Max
HSI-All	100.0	83.5	22.1	672.1
HSI-Threats	100.0	92.1	14.6	788.4
HSI-Acts	100.0	85.8	19.1	700.9
Total articles (000s)	216.9	82.4	91.6	463.1

Note: All HSI series normalized to mean = 100 over the full sample. Total articles in thousands.

The correlation between HSI-Acts and HSI-Threats over the full sample is approximately 0.87, reflecting the tendency for acute violence and elevated threat perceptions to co-occur. However, this masks important dynamics: during calm periods, threats and acts move in tandem; during the post-2017 period, threats persistently exceed acts.

Annual Averages and Key Episodes

Table 2 reports annual averages for all three sub-indices. The data reveal a clear episodic structure with five identifiable high-insecurity regimes, separated by periods of relative calm in 1998–1999, 2007–2009, and 2011–2017.

Table 2: Annual Averages of HSI Sub-Indices, 1997–2025

Year	HSI-All	HSI-Threats	HSI-Acts	Threats/Acts
1997	86.8	75.8	81.6	0.93
1998	44.8	36.6	41.5	0.88
1999	66.7	58.4	63.9	0.91
2000	84.6	86.0	78.2	1.10
2001	67.1	59.2	63.9	0.93
2002	79.5	73.2	74.8	0.98
2003	69.6	62.8	67.8	0.93
2004	219.1	213.6	207.7	1.03
2005	120.8	99.8	114.0	0.88
2006	113.3	106.3	106.1	1.00
2007	68.6	58.7	61.3	0.96
2008	88.0	79.6	86.7	0.92
2009	59.1	46.0	53.4	0.86
2010	112.9	100.8	105.4	0.96
2011	53.6	46.1	48.4	0.95
2012	35.5	29.5	28.3	1.04
2013	44.5	38.6	39.1	0.99
2014	44.4	39.3	40.9	0.96
2015	44.7	40.7	37.2	1.09
2016	51.0	51.7	39.3	1.32
2017	40.5	37.3	32.2	1.16
2018	78.4	73.5	60.3	1.22
2019	75.7	79.5	61.1	1.30
2020	101.0	114.5	83.7	1.37
2021	231.2	214.4	200.1	1.07
2022	143.5	140.5	122.1	1.15
2023	214.6	203.1	144.1	1.41
2024	202.7	225.6	176.5	1.28
2025*	152.7	183.2	133.7	1.37

Note: *2025 covers January–November only. All series normalized to mean = 100. Threats/Acts ratio = HSI-Threats ÷ HSI-Acts.

Major Security Episodes

Table 3 summarises HSI readings for the five most prominent security events in the sample.

Table 3: HSI Readings at Major Security Events

Period	Event	HSI-All	HSI-Thr.	HSI-Acts
Feb–Mar 2004	Aristide departure; MINUSTAH deployment	219.1†	213.6	207.7
Jan–Feb 2010	M7.0 earthquake; humanitarian crisis	112.9†	100.8	105.4
Jul 2021	Assassination of President Moïse	613.5	494.6	669.5
Sep 2021	M7.2 earthquake (southern Haiti)	520.3	493.0	374.6
2023–2024	Gang consolidation; port/airport seizures	208.7†	214.4	160.3
Mar 2024	G9-led assault on Port-au-Prince	501.0	601.2	477.7

Note: †Annual or two-year average; all other figures are monthly readings. Sources: Factiva; IMF staff calculations.

The 2004 political crisis. The removal of President Aristide in February 2004 triggered the most severe sustained insecurity episode in the sample, with an annual average HSI-All of 219 and a monthly peak of 657 in March 2004—the highest reading on record. Both acts and threats spiked simultaneously, consistent with a period of active armed conflict combined with acute uncertainty about the political transition.

The 2010 earthquake. The January 12, 2010 earthquake drove a large but transient HSI spike (annual average 113). The absence of a security crisis comparable to 2004 or 2021 in earthquake-year readings reflects the fundamentally humanitarian—rather than conflict-driven—nature of the episode.

The 2021 dual shocks. The assassination of President Jovenel Moïse on July 7, 2021 produced the highest single-month HSI-Acts reading in the sample (669.5). The subsequent August 2021 earthquake in southern Haiti contributed a second spike, captured primarily through HSI-All (September reading: 520). The 2021 annual average of 231 is the highest on record, exceeding even 2004.

The 2022–2025 gang consolidation period. Unlike earlier crises, this period is characterised by a persistent, elevated baseline rather than discrete high-salience events. Annual averages of 143–215 reflect progressive gang territorial consolidation. HSI-Threats systematically exceeds HSI-Acts (225.6 vs. 176.5 in 2024), consistent with an environment dominated by intimidation and preemptive behavioural adjustments rather than continuous direct confrontation.

March 2024: Gang assault on Port-au-Prince. The coordinated attacks on police stations, the main port, and the international airport produced a monthly HSI-All of 501 and HSI-Threats of 601—the highest threats reading since 2004. Threats exceed acts in this episode (601 vs. 478), reflecting the combination of violence and the existential threat of potential state collapse.

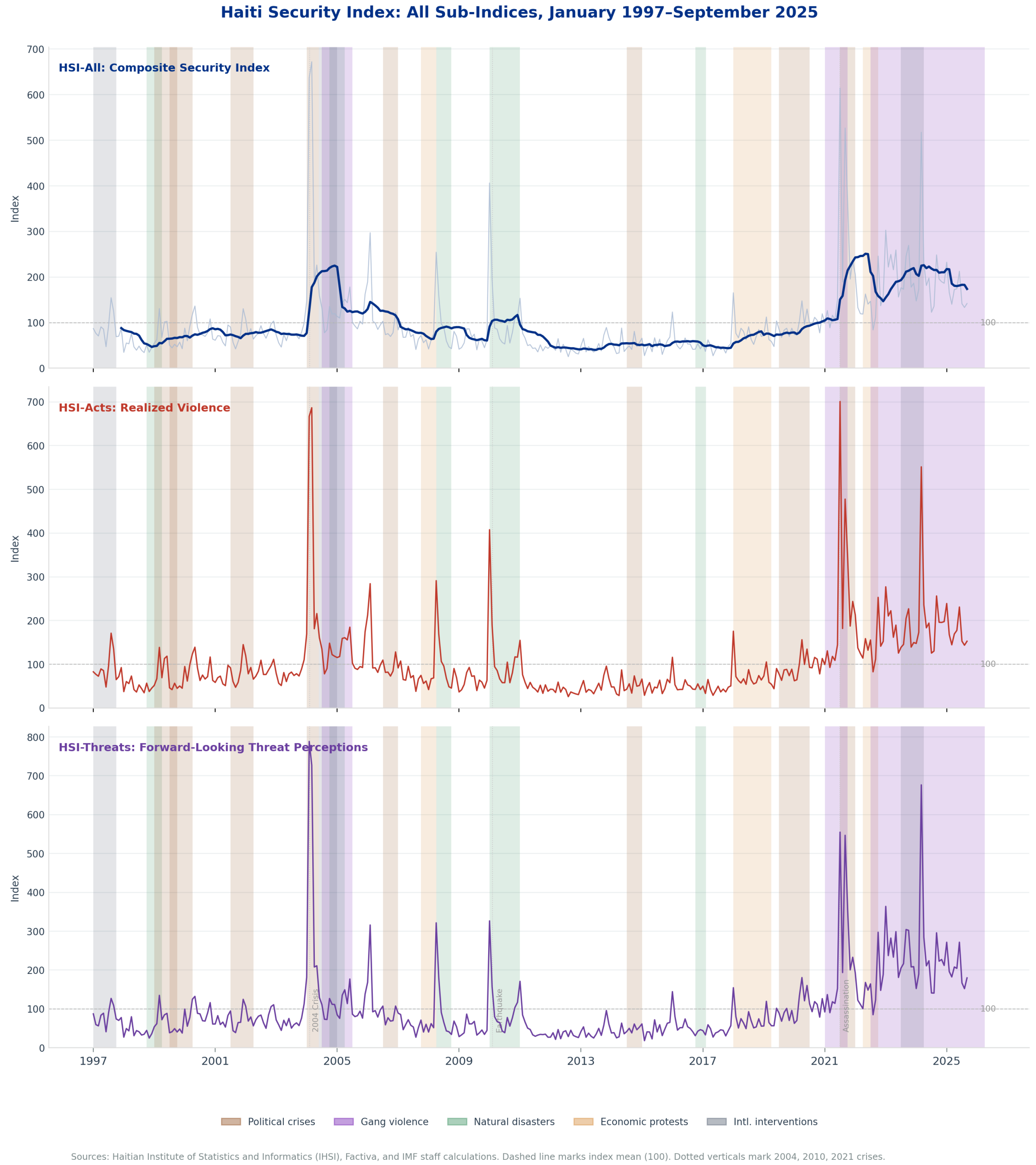

Figure 3: Haiti Security Index: All Three Sub-Indices, January 1997–September 2025

Sources: Haitian Institute of Statistics and Informatics (IHSI), Factiva, and IMF staff calculations. *Note:* Each panel shows a separate sub-index normalized to mean = 100 over the full sample. The dashed horizontal line marks the index mean. Dotted verticals mark the 2004 political crisis, 2010 earthquake, and 2021 presidential assassination. Shaded regions indicate event categories (see legend).

Structural Shift Post-2017: A Threat-Dominated Environment

A key empirical finding is a structural change in the acts–threats relationship after 2017. Over the pre-2017 sample, the average HSI-Threats/HSI-Acts ratio was 0.98—threats and acts were roughly commensurate. After 2017, this ratio rises to 1.28, indicating that threat perceptions systematically and persistently exceed realized violence.

This shift has concrete implications for macroeconomic transmission. When threats exceed acts, the primary channel of insecurity operates through expectations: firms and households respond to perceived risk by adjusting prices, seeking FX, and reducing investment even in the absence of immediate violence. The structural shift is consistent with a transition from episodic, politically motivated violence toward persistent territorial intimidation by gang federations that extract compliance through the *threat* of violence rather than its constant exercise.

Crucially, this post-2017 inversion differs qualitatively from earlier episodes. In the acute crises of 2004 and 2010, HSI-Acts and HSI-Threats spiked together: violence materialized first and perceptions followed. The expectational channel was active but secondary—it amplified realized shocks rather than driving outcomes independently. In the post-2017 period, HSI-Threats persistently exceeds HSI-Acts even outside of discrete crisis events, suggesting that the expectation channel has become structurally dominant. Economic agents have internalised a baseline level of anticipated insecurity that depresses investment, sustains precautionary dollar demand, and constrains firm activity continuously—not just during acute episodes. This distinction matters for policy: addressing realized violence alone is insufficient if threat perceptions remain elevated, as the economic costs of insecurity can persist or even intensify even when incident counts decline.

Macroeconomic Applications

Surveillance and Program Monitoring

The HSI is well-suited for integration into IMF Article IV assessments and Staff-Monitored Program reviews of Haiti, where the absence of timely macroeconomic data is a recurring constraint on analysis. A monthly HSI provides a real-time characterisation of security conditions that can (i) explain turning points in available macroeconomic data, (ii) inform risk scenarios for growth and fiscal projections, and (iii) provide context for policy recommendations on governance and security-sector reform.

For program monitoring, HSI-Threats provides an early-warning dimension: sustained increases in threat perceptions often precede deterioration in fiscal performance and exchange rate stability by one to three months, as behavioural adjustments by firms and households predate the materialisation of violence.

Nowcasting GDP and Inflation

Security shocks affect GDP through multiple channels: port and corridor blockades reduce trade volumes; firm shutdowns reduce output; displacement reduces labour supply; and confidence effects depress investment. The HSI can serve as a high-frequency indicator in nowcasting frameworks that combine scarce official statistics with alternative data.

Preliminary analysis shows that HSI-All Granger-causes quarterly real GDP growth in Haiti over the available sample.

For inflation, the relevant channel runs primarily through HSI-Threats rather than HSI-Acts: anticipated supply disruptions lead firms to preemptively raise prices before blockades materialise. An increase in HSI-Threats of 100 points (one full sample standard deviation) is associated with a measurable acceleration in domestic food price inflation in subsequent months, consistent with import-dependence of the Haitian food supply and the sensitivity of retail prices to expected port access.

Exchange Rate and External Sector Analysis

Haiti's exchange rate (gourde per USD) responds systematically to security conditions. Episodes of acute insecurity—2004, 2021, 2024—are associated with gourde depreciation and widening parallel market premia, consistent with precautionary dollar demand. The HSI, particularly its Threats component, captures the anticipatory dimension: households and firms shift into dollars before violence escalates, compressing the parallel-to-official rate spread in advance of security deterioration.

An HSI-augmented model of the gourde–dollar rate outperforms fundamentals-based models during the 2018–2025 period, a finding consistent with the structural shift toward threat-dominated insecurity that is not well proxied by realized-violence indicators alone.

Remittance and Diaspora Channels

Haiti's diaspora remittances—representing approximately 20–25 percent of GDP—respond to both catastrophic shocks and persistent security deterioration. The assassination of President Moïse in July 2021 was followed by a temporary increase in remittances (a "disaster remittance" response), but the 2022–2024 period of sustained gang control is associated with shifts in remittance channelling away from formal financial institutions toward informal couriers, as recipients avoid moving through insecure corridors to access formal transfer points. The HSI provides a quantitative counterpart to qualitative field assessments of these behavioural responses.

External Validation

Validation Data: BINUH Security and Crime Statistics

To assess the external validity of the HSI, we compare it with the Security and Crime Statistics dataset published by the United Nations Integrated Office in Haiti (BINUH) through the UN Peace and Security Data Hub. The BINUH dataset provides monthly counts of violent incidents reported to UN field personnel, including total incidents involving human insecurity (*ih_total*), murders (*ih_murder*), lynchings, and kidnappings (*kidnap_total*), covering July 2018 through early 2026. These administrative records are compiled independently of news media and therefore constitute a genuinely external benchmark for the text-based HSI.

The overlap period between the two datasets—July 2018 through September 2025—spans 86 months and encompasses some of the most severe security deterioration in Haiti's modern history: the PetroCaribe protests (2018–2019), the Peyi Lok movement, the 2021 presidential assassination, the collapse of fuel subsidies, and the dramatic escalation of gang territorial control from 2022 onward.

Comparative Advantages of the HSI

Before presenting the validation results, it is useful to make precise the informational advantages the HSI holds relative to the BINUH benchmark. Three dimensions stand out.

Temporal coverage. The BINUH crime statistics begin in July 2018, giving a sample of roughly seven years. The HSI extends from January 1997, covering nearly three decades. This longer horizon captures the 2004 political crisis and Aristide ouster, the MINUSTAH period, the 2010 earthquake, the cholera epidemic, and the political turmoil of the Martelly years—episodes that cannot be analysed at all with BINUH data.

Conceptual breadth. BINUH statistics measure incidents that rise to the threshold of formal UN reporting. The HSI, by contrast, captures the full spectrum of security-relevant discourse: political threats, curfews, state-of-emergency declarations, territorial control by armed groups, and international warnings, in addition to realized acts of violence. The HSI-Threats sub-index in particular tracks forward-looking perceptions that are absent from any incident-count database, and that may matter more for economic behaviour than the violence that actually occurs.

Continuity and timeliness. Administrative incident reports depend on field presence, reporting protocols, and institutional continuity that can deteriorate precisely when security is worst. The HSI, sourced from commercial and archival news databases, is available on a near-real-time basis without gaps.

Visual Comparison

Figure 4 plots the HSI-All series against BINUH total violent incidents and kidnappings over the 2018–2025 overlap period. The two series track each other closely across major turning points. Both measures rise sharply around the 2019 Peyi Lok movement and political paralysis, dip modestly during the pandemic restrictions of 2020, surge following the July 2021 assassination of President Moïse, and then rise steeply and persistently from 2022 onward as gang violence escalated. The HSI tends to lead or coincide with peaks in the BINUH counts, consistent with its role as a perception-based indicator that aggregates forward-looking signals alongside contemporaneous violence.

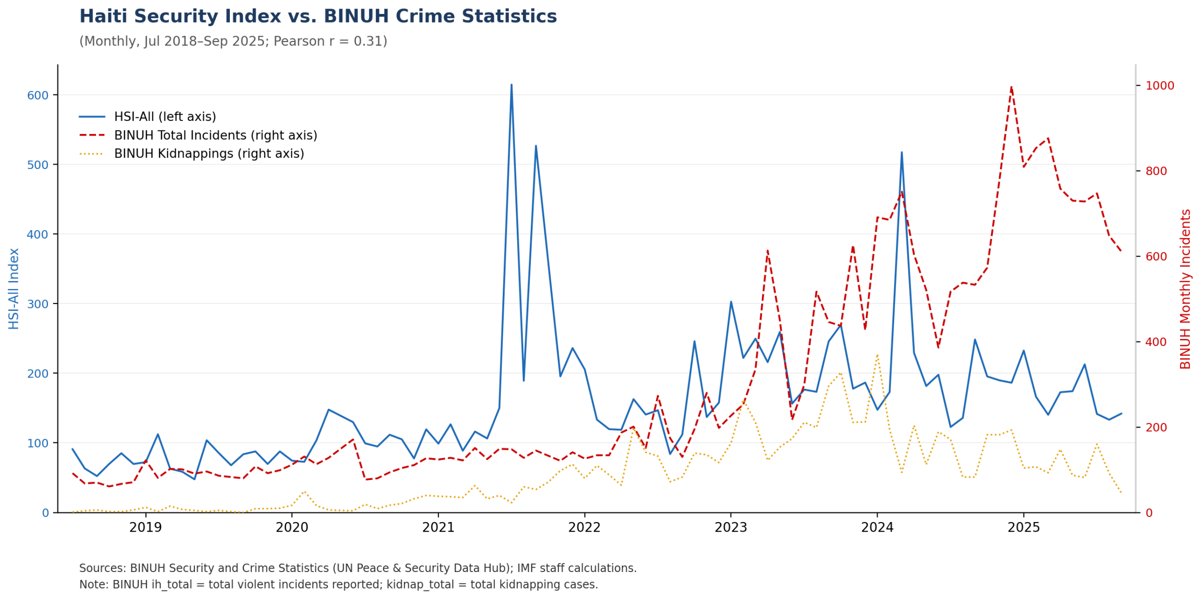

Figure 4: HSI-All vs. BINUH Crime Statistics, July 2018–September 2025
Sources: BINUH Security and Crime Statistics (UN Peace and Security Data Hub); IMF staff calculations. HSI-All shown on left axis; BINUH monthly incident counts on right axis.

Correlation and Regression Analysis

Table 4 reports Pearson and Spearman rank correlations between HSI-All and three BINUH crime indicators over the 86-month overlap sample, together with OLS regression coefficients from univariate regressions of each BINUH indicator on HSI-All.

Table 4: Validation: HSI-All vs. BINUH Crime Indicators, Jul 2018–Sep 2025

	Pearson r		Spearman ρ		OLS	
BINUH Indicator	r	p-value	ρ	p-value	$\hat{\beta}$	R^2
Total incidents	0.311	0.004***	0.704	<0.001***	0.809	0.097
Murders	0.302	0.005***	0.699	<0.001***	0.747	0.091
Kidnappings	0.354	0.001***	0.725	<0.001***	0.303	0.125

Notes: $N = 86$ monthly observations (July 2018–September 2025). OLS regressions: $\text{BINUH}_t = \alpha + \beta \cdot \text{HSI-All}_t + \varepsilon_t$. Pearson r measures linear co-movement; Spearman ρ measures rank co-movement and is more robust to the non-linear escalation pattern evident in 2022–2025. *** $p < 0.01$.

The results support the external validity of the HSI. The Spearman rank correlations are large and highly significant across all three crime indicators: $\rho = 0.704$ for total incidents, $\rho = 0.699$ for murders, and $\rho = 0.725$ for kidnappings (all $p < 0.001$). The Pearson correlations are more modest (0.30–0.35) but remain statistically significant at the 1% level, reflecting the nonlinear acceleration of both series in the 2022–2025 period that rank-based measures capture better than linear correlations.

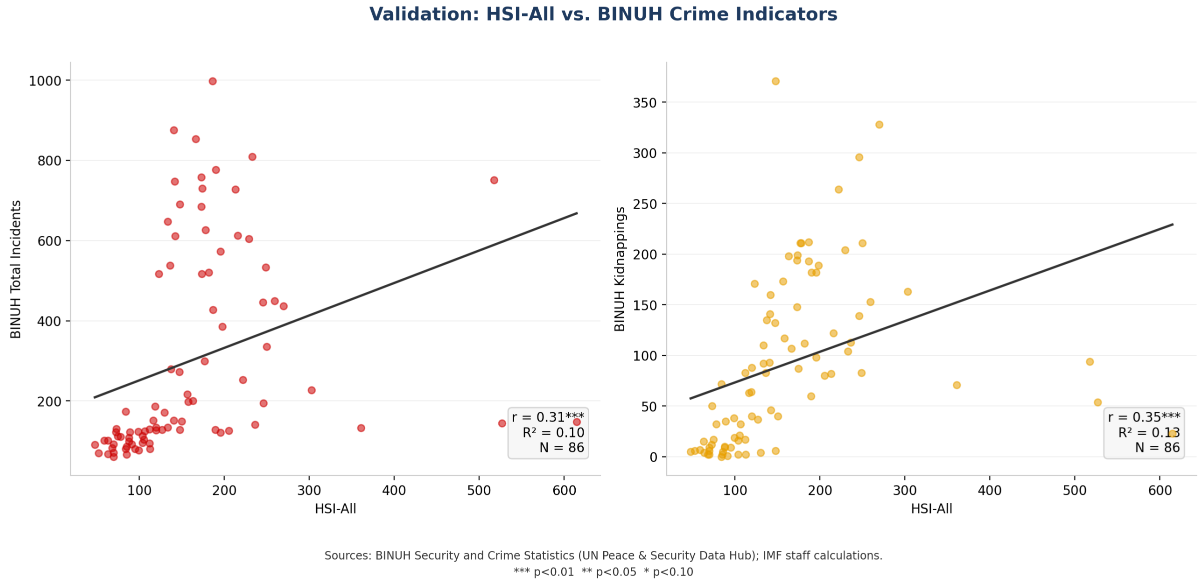

Figure 5: Scatter Plots: HSI-All vs. BINUH Crime Indicators
Sources: BINUH Security and Crime Statistics (UN Peace and Security Data Hub); IMF staff calculations. OLS regression line shown; r = Pearson correlation; *** $p < 0.01$.

Figure 5 presents scatter plots for total incidents and kidnappings. The positive slope is visually evident in both panels, with the regression line capturing the central tendency of the relationship. The scatter around the line reflects the broader conceptual scope of the HSI—which incorporates political threats, institutional signals, and media salience in addition to ground-level incident counts—and the measurement differences between a perception-based indicator and an administrative reporting system.

Discussion

Taken together, the validation results establish that the HSI-All tracks ground-truth crime outcomes with statistically and economically meaningful correlations over the available overlap period. The strong rank correlations in particular indicate that the HSI reliably orders months by security severity—a property directly useful for surveillance and early warning applications. At the same time, the lower Pearson R^2 values (0.09–0.13) underscore that the HSI captures dimensions of insecurity beyond what is recorded in UN incident reports: threat escalation, political risk, and diffuse fear are embedded in the text signal even when they do not generate a reportable incident. This divergence is a feature rather than a limitation, precisely because these latent dimensions of insecurity are the ones most likely to affect macroeconomic behaviour through expectations.

Conclusion

This paper introduces the Haiti Security Index (HSI), a monthly text-based measure of domestic security conditions covering January 1997 to March 2026. The HSI tracks the prominence of security-related news coverage, separately quantifying realized violent acts (HSI-Acts) and forward-looking threat perceptions (HSI-Threats), normalized by total media volume and scaled to a mean of 100. The index successfully captures all major security episodes in Haiti's modern history and reveals a structural shift after 2017 toward a threat-dominated environment—a finding with direct implications for the economic channels through which security affects macroeconomic outcomes.

The HSI fills a critical data gap for a country where standard macroeconomic surveillance is constrained by data availability and timeliness. Its monthly frequency, near-real-time availability, and replicable methodology make it immediately useful for Article IV consultations, SMP reviews, nowcasting, and risk scenario construction.

External validation against BINUH Security and Crime Statistics confirms that the HSI tracks ground-truth violent incidents with strong rank correlations ($\rho \approx 0.70$–0.73, $p < 0.001$) over the 2018–2025 overlap period. The validation also highlights what administrative crime data alone cannot provide: a forward-looking threat signal spanning nearly three decades, continuous and timely, and sensitive to the full spectrum of security-relevant discourse—political instability, institutional fragility, and anticipatory fear—that shapes macroeconomic outcomes well before violence materialises in incident reports.

Several extensions are planned. First, an expanded Spanish-language component will incorporate additional regional media. Second, district-level geographic tagging will enable sub-national security indicators for Haiti's departments. Third, automated pipeline integration with Factiva will enable monthly updates alongside IMF surveillance products.

More broadly, the methodology is directly transferable to other fragile and conflict-affected states where standard macroeconomic data are limited, offering a scalable, low-cost approach to bridging the informational gaps that constrain evidence-based policy and surveillance.

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Keyword Dictionary

Table A1: HSI Keyword Dictionary — Acts, Threats, and Institutional Terms

All searches include bilingual English and French variants and require keywords to appear within 50 words of a Haiti location reference (`NEAR/50`). Exclusion filters remove sports, entertainment, natural disasters, and metaphorical uses.

Table 5: HSI Keyword Dictionary

Index	Subcategory	Example Terms
HSI-Acts	Physical violence	*attack, assault, ambush, raid, strike, clash, fight, battle, shoot*, gunfire, firefight, crossfire, brawl, skirmish*
HSI-Acts	Killings & casualties	*kill*, murder*, homicide, massacre, execute*, lynch*, death*, fatalit*, injur*, wound*, casualt*, victim**
HSI-Acts	Explosives & terrorism	*bomb*, blast, grenade, explosion, explosive, detonat*, terror*, suicide bomb, terrorist attack*
HSI-Acts	Kidnapping & coercion	*kidnap*, abduct*, hostage, ransom, hijack*, extort*, blackmail**
HSI-Acts	Riots & disorder	*riot*, loot*, arson, burn*, fire set, vandal*, mob attack, crowd attack*
HSI-Acts	Blockades & sabotage	*blockade, roadblock, barricade, sabotag*, cut off, shutdown*
HSI-Threats	Fear & risk language	*threat*, warn*, fear*, risk*, concern*, danger*, menace*, panic, anxiety, worry, tense*
HSI-Threats	Anticipatory crisis framing	*crisis, troubl*, disput*, tension*, instabilit*, unrest, insecurity, under siege, heightened alert*
HSI-Threats	Policy & emergency context	*state of emergency, curfew, possible violence, potential attack, civil unrest, calls for calm*
HSI-Threats	Imminence language	*imminen*, inevitable, risk of attack, risk of kidnapping, fear of violence*
Institutional	Domestic security	*police, haitian national police, PNH, security force*, authority, government, ministry*

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Index	Subcategory	Example Terms
Institutional	Non-state actors	*gang*, armed group*, militia*, vigilante**
Institutional	International presence	*UN mission, peacekeeper*, multinational security support, MSS, foreign troop*, intervention*

Table A2: News Sources and Coverage

Table 6: News Sources and Coverage

Outlet	Lang.	Type	Since	Role
Le Nouvelliste	French	Local	1997	Primary anchor
Gazette Haiti	French	Local	1997	Primary anchor
Haiti Libre	Fr/En	Local	1997	Primary anchor
Haitian Times	English	Diaspora	1997	Primary anchor
AFP / Reuters / AP	English	Wire	1977	International coverage
CE NoticiasFinancieras	Spanish	Reg. wire	1997	Regional context
Miami Herald	English	Print	1997	Caribbean desk
BBC / NYT / WaPo / CP	English	Intl	1997	Global reach

Note: All sources accessed via Factiva. “Since” is the first year with consistent monthly observations.